

5	(a)	First law of thermodynamics states that the increase in internal energy of a gas is the sum of the thermal energy supplied to the gas and the work done on the gas.	A1
	(b) (i)	Using the ideal gas equation, $P = \frac{nRT}{V}$ $P_i = \frac{nRT_i}{V_i} = \frac{nR(21.0 + 273.15)}{3.49 \times 10^{-6}} = nR(8.43 \times 10^4)$ $P_f = \frac{nRT_f}{V_f} = \frac{nR(53.0 + 273.15)}{3.87 \times 10^{-6}} = nR(8.43 \times 10^4)$ Since the mass of the gas (or number of moles) is constant, the final and initial pressures are the same. Or Since $PV=nRT$, and the number of moles is the same, the ratio of V/T must be constant. $\frac{V_1}{T_1} = \frac{3.49 \times 10^3 \times 10^{-6}}{21.0 + 273.15} = 1.19 \times 10^{-5}$ $\frac{V_2}{T_2} = \frac{3.87 \times 10^3 \times 10^{-6}}{53.0 + 273.15} = 1.19 \times 10^{-5}$ the pressure is also constant	B1 M1 A0
	(b) (ii)	Since pressure is constant, $W_{by} = p\Delta V$ $= 4.2 \times 10^5 \times (3.87 - 3.49) \times 10^3 \times 10^{-6}$ $= 160 \text{ J}$ (1 mark for students whose answer is – 160 J)	C1 A1
	(b) (iii)	$\Delta U = Q_{to} + W_{on}$ $= (+ 565) + (- 160)$ $= 405 \text{ J}$	C1 A1
	(b) (iv)	The internal energy of a real gas is defined as the sum of the microscopic kinetic energy and potential energy of the gas molecules. However, for an ideal gas, it is assumed that there are no intermolecular forces of attraction between the gas molecules. This means that the gas molecules do not possess microscopic potential energy. Hence, the the change in its internal energy is totally due to the change in the total microscopic kinetic energies of the gas molecules.	B1 B1 B1 A0

C	(c)	Magnetic flux density is the force acting per unit current per unit length on a	D1
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